

AP CSP Vocabulary Glossary

Working in Python · alphabetical · 144 terms · AP-tested terms only

Terms marked ^{AP} appear on the College Board AP CSP exam's own vocabulary list. A superscript chapter number links to where this book defines the term.

Vocabulary drawn from *Working in Python*, a fork of Allen Downey's *Think Python*, 3rd edition (CC BY-NC-SA 4.0), and this course's own AP CSP vocabulary reference. AP CSP definitions here are written in this course's own words, not copied from the College Board's Course and Exam Description.

A

abstraction^{AP} Reducing complexity by exposing essential features and hiding implementation detail.

aggregation of information^{AP} Combining separately harmless pieces of data to identify or profile a person.

algorithm^{AP} A finite sequence of steps that solves a problem or completes a task. Can be written in English, pseudocode, or code.

algorithmic efficiency^{AP} An estimate of the computational resources an algorithm uses as a function of input size.

analog data^{AP B} Values that vary continuously and smoothly, with no steps.

anonymization^{AP} Removing identifying details from data — and its limits, since aggregation can re-identify.

API^{AP} The specification of how a library's procedures are called and how they behave.

argument^{AP 2} A value provided to a function when the function is called.

ASCII^{AP B} A table assigning a number from 0 to 127 to each of a small set of characters.

assignment statement^{AP 2} A statement that assigns a value to a variable.

authentication^{AP} Verifying that a user is who they claim to be.

B

bandwidth^{AP} Maximum data transmittable per unit time, in bits per second.

beneficial and harmful effects^{AP} Every computing innovation has both, and they can fall on different groups of people.

bias^{AP} Systematic unfairness embedded in an algorithm or in the data it was built from.

binary^{AP B} Base-2 representation, using only the digits 0 and 1.

binary search^{AP} Repeatedly halving a sorted list to locate a value.

bit^{AP B} A single binary digit, 0 or 1.

boolean expression^{AP 5} An expression whose value is either True or False.

byte^{AP B} Eight bits. Enough to hold one of 256 values.

C

certificate authority^{AP} An organization that issues and vouches for digital certificates.

citizen science^{AP} Scientific research carried out partly by volunteers using their own devices.

code segment^{AP} A portion of a program, one or more lines, treated as a unit.

collaboration^{AP} Working with others to develop a computing innovation; benefits from varied perspectives and skills.

comment^{AP 2} Text included in a program that provides information about the program but has no effect on its execution.

compound conditional^{AP} A single condition combining multiple Boolean expressions with AND, OR, or NOT.

computer network^{AP} Interconnected computing devices able to send and receive data.

computer virus^{AP} Malware that copies itself, typically by attaching to other programs or files.

computing device^{AP} A physical artifact that can run a program.

computing innovation^{AP} A product, service, or concept that includes a program as an integral part of its function. May be physical, software, or conceptual.

computing system^{AP} A group of computing devices and programs working together.

concatenation^{AP 1} Joining two strings end-to-end.

conditional statement^{AP 5} A statement that controls the flow of execution depending on some condition. Informally, this is usually an if statement, which might include an elif and an else. (*AP calls this selection.*)

constant^{AP} A named value that does not change during execution.

cookies^{AP} Small data files stored by a site to track a user's state or behavior.

copyright^{AP} The exclusive right of a creator to reproduce, distribute, and display a work.

correlation^{AP} An association between two variables where changes in one accompany changes in the other.

creative commons^{AP} A set of public licenses letting a creator pre-authorize specific reuse under stated conditions.

crowdsourcing ^{AP} Gathering input, funding, or labor from many people over the Internet.

D

data ^{AP} Values that can be stored and processed by a program.

data abstraction ^{AP} Using a data structure, typically a list, to represent related values as one named unit, hiding the individual pieces.

data cleaning ^{AP} Making data uniform and consistent without changing its meaning.

data mining ^{AP} Analyzing large data sets to find patterns and relationships.

data set ^{AP} A collection of related data organized for analysis.

debugging ^{AP 1} The process of finding and correcting errors.

decidable problem ^{AP} A yes/no problem for which an algorithm can always produce the correct answer.

decimal ^{AP B} Base-10, the system you already use.

digital certificate ^{AP} A credential issued by an authority binding a public key to an identity.

digital data ^{AP B} Values represented in discrete steps, ultimately as bits.

digital divide ^{AP} Unequal access to computing and the Internet across socioeconomic, geographic, or demographic lines.

distributed computing ^{AP} Multiple devices in different locations running parts of one program.

DNS ^{AP} Translates domain names into IP addresses.

docstring ^{AP 4.A} A string at the beginning of a function that documents what the function does; unlike a comment, it is stored on the function and can be read by the program. (*AP calls this program documentation.*)

E

element ^{AP 9} One of the values in a list or other sequence.

encryption ^{AP} Encoding data so only holders of the key can read it.

event ^{AP} An action supplied to a program as input: key press, click, sensor reading.

event-driven programming ^{AP} Statements execute in response to events rather than in top-to-bottom sequence.

expression ^{AP 1} A combination of variables, values, and operators.

F

fault tolerance ^{AP} Continuing to operate when components fail.

filtering ^{AP 10} Looping through a sequence and selecting or omitting elements. (*AP calls this data filtering.*)

function ^{AP 1} A named sequence of statements that performs some useful operation. Functions may or may not take arguments and may or may not produce a result. (*AP calls this a procedure.*)

H

hand tracing ^{AP A} Working through code on paper line by line, writing down each variable's value, in order to find an error without running the program.

heuristic ^{AP} An approach that finds a good-enough solution when finding the optimal one is impractical.

hexadecimal ^{AP B} Base-16, using 0 through 9 and A through F. Four bits per digit, so one byte is exactly two hex digits.

HTTP / HTTPS ^{AP} The Web's request/response protocol; HTTPS adds encryption in transit.

I

incremental development ^{AP 6} A program development plan intended to avoid debugging by adding and testing only a small amount of code at a time.

index ^{AP 8} An integer value used to select an item in a sequence, such as a character in a string. In Python indices start from 0.

infinite loop ^{AP} A loop whose ending condition never becomes true.

information ^{AP} Meaning, patterns, or insight extracted from data.

integer ^{AP 1} A type that represents numbers with no fractional or decimal part.

intellectual property ^{AP} Creative or technical work legally owned by its creator.

interface design ^{AP 4} A process for designing the interface of a function, which includes the parameters it should take. (*AP calls this procedural abstraction.*)

internet ^{AP} A network of interconnected networks using standardized open protocols.

IP ^{AP} Protocol for addressing devices and routing packets between them.

IP address ^{AP} The unique identifier assigned to a device on a network.

iterative development ^{AP} Repeated cycles of build, feedback, and revision; earlier phases get revisited.

K

keylogging ^{AP} Recording keystrokes to capture passwords and confidential input.

L

linear search ^{AP 7} A computational pattern that searches through a sequence of elements and stops when it finds what it is looking for.

list ^{AP 9} An object that contains a sequence of values. (*on the AP exam, list indices start at 1; in Python, at 0.*)

logical operator^{AP 5} One of the operators that combines boolean expressions, including and, or, and not.

loop^{AP 3} A statement that runs one or more statements, often repeatedly. (*AP calls this iteration.*)

lossless compression^{AP B} Reduces size while allowing the original to be reconstructed exactly.

lossy compression^{AP B} Reduces size further, but only an approximation of the original can be recovered.

M

machine learning^{AP} Systems that improve at a task from data rather than from explicit programming.

malware^{AP} Software intended to damage or take control of a system.

metadata^{AP} Data describing other data: creation date, size, author, location, format.

modularity^{AP} Dividing a program into independent parts each responsible for one aspect.

module^{AP 2} A file that contains Python code, including function definitions and sometimes other statements. (*AP calls this a software library.*)

modulus operator^{AP 5} An operator, %, that works on integers and returns the remainder when one number is divided by another. (*AP calls this MOD.*)

multifactor authentication^{AP} Requiring evidence from two or more categories: knowledge, possession, inherence.

N

nested conditional^{AP 5} A conditional statement that appears in one of the branches of another conditional statement.

O

open access^{AP} Research or data made available free of access restrictions.

open source^{AP} Software whose source is public and may be modified and redistributed.

open standard^{AP} A publicly available specification that anyone may implement without permission or fee.

overflow error^{AP B} An error that happens when a value is too large for the number of bits available to hold it.

P

packet^{AP} A chunk of data carrying metadata such as its destination, sent across a network.

packet switching^{AP} Splitting a message into packets that travel independently, possibly by different routes, and are reassembled at the destination.

pair programming^{AP} Two programmers at one workstation: one drives, one reviews and thinks ahead.

parallel computing^{AP} A program split into operations, some performed simultaneously.

parameter^{AP 3} A name used inside a function to refer to the value passed as an argument.

phishing^{AP} Tricking a user into revealing private information by impersonating a trusted party.

PII^{AP} Information that identifies or can be linked to a specific individual.

pixel^{AP} The smallest addressable color element of a display or image.

plagiarism^{AP} Presenting someone else's work as your own.

procedure^{AP A} The exam's word for a named, reusable block of code, whether or not it returns a value. This book says function. In older languages the two words were distinct: a procedure returned nothing, a function returned a value.

program^{AP} A collection of statements that performs a task when run.

program function^{AP A} What a program does when it runs, described as behavior.

program input^{AP A} Data a program receives while it is running.

program output^{AP A} What a program produces: displayed text, a returned value, a file, a sound, a movement.

program purpose^{AP A} The need a program serves, or the problem it solves; why it exists.

protocol^{AP} An agreed set of rules governing system behavior.

pseudorandom^{AP 12} A pseudorandom sequence of numbers appears to be random, but is generated by a deterministic program. (*AP calls this RANDOM(a, b) — inclusive on both ends, unlike Python's range.*)

public key encryption^{AP} A public key encrypts; a different private key decrypts.

R

reasonable time^{AP} Run time that grows polynomially or slower with input size.

redundancy^{AP} Duplicate paths or components that allow operation to continue after a failure.

relational operator^{AP 5} One of the operators that compares its operands: ==, !=, >, <, >=, and <=.

return value^{AP 6} The result of a function. If a function call is used as an expression, the return value is the value of the expression. (*AP calls this the RETURN statement.*)

RGB^{AP B} Color stored as three numbers, the amounts of red, green, and blue, each usually one byte.

rogue access point^{AP} An unauthorized wireless access point used to intercept network traffic.

roundoff error^{AP A,B} A loss of precision that happens because a fixed number of bits cannot represent some numbers exactly.

router^{AP} A device that forwards packets between networks toward their destination.

routing^{AP} Determining a path from sender to receiver across a network.

runtime error^{AP 2} An error that causes a program to display an error message and exit.

S

sampling^{AP B} Approximating an analog signal by measuring it at regular intervals. Two independent settings: how often you measure (rate) and how precisely you record each measurement (bit depth).

scalability^{AP} A system's ability to keep working acceptably as demand grows.

semantic error^{AP 2, A} An error that lets the program run but produces a wrong result. (*AP calls this a logic error.*)

sequencing^{AP} Executing steps in the order written.

sequential computing^{AP} Operations performed one at a time, in order.

simulation^{AP} An abstraction of a real phenomenon using varying values to represent changing states, run for a purpose.

speedup^{AP} Sequential time divided by parallel time.

string^{AP 1} A type that represents sequences of characters.

substring^{AP} A contiguous portion of a string.

symmetric key encryption^{AP} One shared key both encrypts and decrypts.

syntax error^{AP 1} An error in a program that makes it impossible to parse — and therefore impossible to run.

T

TCP^{AP} Transport protocol providing reliable, ordered delivery with retransmission.

test case^{AP A} A specific input paired with its expected output. Good sets include typical, boundary, and edge values.

testing^{AP A} Checking that a program behaves correctly by running it on chosen inputs and comparing what comes out against what should have come out.

traversal^{AP} Iterating over the items of a list. Full traversal visits every element; partial stops early.

type^{AP 1} A category of values. The types we have seen so far are integers (type int), floating-point numbers (type float), and strings (type str). (*AP calls this data type.*)

U

UDP^{AP} Lightweight transport protocol with minimal error checking; faster, less reliable.

undecidable problem^{AP} A problem for which no algorithm can always give a correct yes/no answer for every input.

Unicode^{AP B} A far larger table, covering the writing systems ASCII left out.

unreasonable time^{AP} Run time that grows exponentially or factorially with input size.

V

variable^{AP 2} A name that refers to a value.

W

world wide web^{AP} A system of linked pages and files that runs on top of the Internet using HTTP.